• INDETERMINANT FORM & MAXIMA & MINIMA & MEAN VALUE THEOREM (LAGRANGE, Cauchy)

PYQ → 2022 [Marks → 7]

Q.2.(b) Use L'Hospital rule to find the following limits.

$$\lim_{x \to 0} \frac{\cos x - \log(1+x) - 1 + x}{\sin^2 x}$$

⇒ Given;

$$\lim_{x \to 0} \frac{\cos x - \log(1+x) - 1 + x}{\sin^2 x} \quad \left[\frac{0}{0} \text{ form}\right]$$

∴ applying L'Hospital's rule;

$$\lim_{x \to 0} \frac{-\sin x - \frac{1}{1+x} + 1}{2 \sin x \cos x} \quad \left[\frac{0}{0} \text{ form}\right]$$

applying L'Hospital's rule;

$$\lim_{x \to 0} \frac{-\cos x + \frac{1}{(1+x)^2}}{2\left[\cos^2 x - \sin^2 x\right]}$$

$$= 0$$

∴ Limit of the following question is 0

Expected Question (2023-24);

Q.1. Evaluate

$$\lim_{x\to 0} \frac{e^x \sin x - x - x^2}{x^2 + x\log(1-x)}$$

⇒ Given;

$$\lim_{x\to 0} \frac{e^x \sin x - x - x^2}{x^2 + x\log(1-x)} \quad \left[\frac{0}{0} \text{ form}\right]$$

applying L'Hospitals ~~form~~ rule;

$$\lim_{x\to 0} \frac{e^x \sin x + e^x \cos x - 2x - 1}{2x + \log(1-x) - \frac{x}{1-x}} \quad \left[\frac{0}{0} \text{ form}\right]$$

applying L'Hospitals rule;

$$\lim_{x\to 0} \frac{e^x \sin x + e^x \cos x + e^x \cos x - e^x \sin x - 2}{\frac{-1}{1-x} - \left[\frac{(1-x)+x}{(1-x)^2}\right] + 2}$$

$$\lim_{x\to 0} \frac{2e^x \cos x - 2}{\frac{-1}{1-x} - \frac{1}{(1-x)^2} + 2} \quad \left[\frac{0}{0} \text{ form}\right]$$

$$\lim_{x\to 0} \frac{2\left[e^x \cos x - e^x \sin x\right]}{\frac{-1}{(1-x)^2} - \frac{1}{(1-x)^3} \times (2)} \quad \left[e^0 = 1\right]$$

$$= \frac{2\left[e^0 \cos 0 - e^0 \sin 0\right]}{\frac{-1}{(1-0)^2} - \frac{2}{(1-0)^3}} = -\frac{2}{3}$$

Q.2. $\underset{x \to 0}{Lt} \dfrac{xe^x - \log(1+x)}{x^2}$

$\Rightarrow$ Given;

$$\underset{x \to 0}{Lt} \frac{xe^x - \log(1+x)}{x^2} \quad \left[\frac{0}{0} \text{ form}\right]$$

applying L'Hospitals rule;

$$= \underset{x \to 0}{Lt} \frac{e^x + xe^x - \frac{1}{1+x}}{2x} \quad \left[\frac{0}{0} \text{ form}\right]$$

applying L'Hospital's rule;

$$= \underset{x \to 0}{Lt} \frac{e^x + e^x + xe^x + \frac{1}{(1+x)^2}}{2}$$

$$= \frac{2e^0 + 0e^0 + \frac{1}{(1+0)^2}}{2} \quad [e^0 = 1]$$

$$= \frac{3}{2}$$

$\therefore$ Here, limit of the following question is $\frac{3}{2}$

Q.2. $\underset{x \to 0}{Lt} \dfrac{e^x - e^{\sin x}}{x - \sin x}$

$\Rightarrow$ Given;

$$\underset{x \to 0}{Lt} \frac{e^x - e^{\sin x}}{x - \sin x} \quad \left[\frac{0}{0} \text{ form}\right]$$

applying L'Hospital's rule;

$$\underset{x \to 0}{Lt} \frac{e^x - e^{\sin x} \cdot \cos x}{1 - \cos x} \quad \left[\frac{0}{0} \text{ form}\right]$$

applying L'Hospitals rule;

$$\underset{x \to 0}{Lt} \frac{e^x - \left[e^{\sin x}[-\sin x] + \cos^2 x e^{\sin x}\right]}{+\sin x} \quad \left[\frac{0}{0} \text{ form}\right]$$

applying L'Hospital's rule;

$$\underset{x \to 0}{Lt} \frac{e^x - \left[-\left[\sin x \cos x e^{\sin x} + \cos x e^{\sin x}\right] + 2\cos x(-\sin x) \cdot e^{\sin x} + e^{\sin x} \cos^3 x\right]}{\cos x}$$

$$= \frac{e^0 + \left[\sin 0 \cos 0 e^{\sin 0} + \cos 0 e^{\sin 0}\right] + 2\cos 0 \sin 0 e^{\sin 0} - e^{\sin 0} \cos^3 0}{\cos 0}$$

$$= \frac{1 + 1 - 1}{1} = 1$$

Hence, limit of the following question is 1.

Q.4. $\underset{x \to 0}{Lt} \dfrac{\sin\{\log(1+x)\}}{\log(1+\sin x)}$

$\Rightarrow$ Given;

$$\underset{x \to 0}{Lt} \frac{\sin\{\log(1+x)\}}{\log(1+\sin x)} \quad \left[\frac{0}{0} \text{ form}\right] \Rightarrow$$

$$\underset{x \to 0}{Lt} \frac{\dfrac{\cos\{\log(1+x)\}}{1+x}}{\dfrac{\cos x}{1+\sin x}}$$

$$\underset{x \to 0}{Lt} \frac{\cos\{\log(1+x)\}\,(1+\sin x)}{(\cos x)(1+x)}$$

$$\frac{\cos\{\log(1+0)\}\,(1+\sin 0)}{(\cos 0)\,(1+0)}$$

$$\frac{(\cos 0)\,1}{1\,(1)} = 1$$

Hence, limit of this question is 1

Q.5 If $\lim_{x \to 0} \frac{\sin 2x + a \sin x}{x^3}$ is finite, find the value of a and the limit.

V.V.I

→ Given;

$$\lim_{x \to 0} \frac{\sin 2x + a \sin x}{x^3}$$

Now; LHS;

$$\lim_{x \to 0} \frac{\sin 2x + a \sin x}{x^3} \quad \left[\frac{0}{0} \text{ form}\right]$$

applying L'Hospital's rule;

$$\lim_{x \to 0} \frac{2 \cos 2x + a \cos x}{3x^2}$$

Numerator ~~Denominator~~ should be zero to ~~make~~ be in $\left[\frac{0}{0} \text{ form}\right]$

after taking limit

Numerator ~~Denominator~~ $\Rightarrow 2\cos 2 \cdot 0 + a \cos 0 = 0$

$$2 + a = 0$$

$$a = -2$$

Here, $\lim_{x \to 0} \frac{2\cos 2x - 2\cos x}{3x^2} \quad \left[\frac{0}{0} \text{ form}\right]$

apply L'Hopital's rule;

$$\lim_{x \to 0} \frac{-4 \sin 2x + 2 \sin x}{6x} \quad \left[\frac{0}{0} \text{ form}\right]$$

applying L' hospital's rule;

$$\lim_{x \to 0} \frac{-8 \cos 2x + 2\cos x}{6}$$

$$\frac{-8\cos 0 + 2\cos 0}{6}$$

$$= \frac{-6}{6} = -1$$

Hence the value of $a = -2$ and the limit is -1

~~Q6. Find a, b if $\lim_{x \to 0} \frac{a\sin 2x + b\sin x}{x^3} = \frac{5}{2}$~~

~~⇒ Given;~~

~~$\lim_{x \to 0} \frac{a\sin x + b\sin x}{x^3}$~~

Q.6. Find a, b, c so that

V.V.I

$$\lim_{x \to 0} \frac{ae^{x} - b\cos x + ce^{-x}}{x \sin x} = 2$$

$\Rightarrow$ Given;

$$\lim_{x \to 0} \frac{ae^{x} - b\cos x + ce^{-x}}{x \sin x} = 2$$

Now, LHS;

$$\lim_{x \to 0} \frac{ae^{x} - b\cos x + ce^{-x}}{x \sin x}$$

Numerator should be zero to be in form of $\frac{0}{0}$.

after taking limit

Numerator $\Rightarrow$ $ae^{0} - b\cos 0 + ce^{-0} = 0$

$a - b + c = 0$ —— (i)

Here, $\lim_{x \to 0} \frac{ae^{x} - b\cos x + ce^{-x}}{x \sin x}$ $\left[\frac{0}{0} \text{ form}\right]$

applying L' hospital's rule;

$$\lim_{x \to 0} \frac{ae^{x} + b\sin x - ce^{-x}}{x\cos x + \sin x}$$

again, Numerator should be zero to be in $\left[\frac{0}{0} \text{ form}\right]$

after taking limit, Numerator be;

$ae^0 + b\sin 0 - ce^{-0} = 0$

$a - c = 0$ ——— (ii)

Now, $\lim_{x \to 0} \frac{ae^x + b\sin x - ce^{-x}}{x\cos x + \sin x}$ [$\frac{0}{0}$ form

applying L'hospital's rule;

$\lim_{x \to 0} \frac{ae^x + b\cos x + ce^{-x}}{\cos x - x\sin x + \cos x}$

$\frac{ae^0 + b\cos 0 + ce^{-0}}{2\cos 0 - 0\sin 0} = 2$

$\frac{a + b + c}{2} = 2$

$a + b + c = 4$ ——— (iii)

from eq (i), (ii), and (iii),

$a - b + c = 0$

$a - c = 0$

$a + b + c = 4$

from eq - (ii)

$a = c$

put the value $a=c$ in eq - (i)

$c - b + c = 0$
$2c = b$

put the value $a=c$ and $b=2c$ in eq-(iii)

$c + 2c + c = 4$
$4c = 4$
$c = 1$

put $c=1$ in eq — (2)

$a - 1 = 0$
$a = 1$

put $c=a=1$ in eq - (1)

$1 - b + 1 = 0$
$b = 2$

~~Hea~~ Hence the value of

$a = 1$
$b = 2$
$c = 1$

Q. 7. Find $\lim_{x \to \pi/2} (\sin x)^{\tan x}$

$\Rightarrow$ Given;

$$\lim_{x \to \pi/2} (\sin x)^{\tan x} \qquad [1^{\infty} \text{ form}]$$

let

$$z = \lim_{x \to \pi/2} (\sin x)^{\tan x}$$

$$\log z = \lim_{x \to \pi/2} \log (\sin x)^{\tan x}$$

$$\log z = \lim_{x \to \pi/2} \tan x \log (\sin x)$$

$$\log z = \lim_{x \to \pi/2} \frac{\log \sin x}{\cot x} \qquad \left[\frac{0}{0} \text{ form}\right]$$

apply L' hospital's rule

$$\log z = \lim_{x \to \pi/2} \frac{\frac{1}{\sin x} \cdot \cos x}{-\operatorname{cosec}^2 x}$$

$$\log z = -\lim_{x \to \pi/2} \frac{\cos x}{\sin x} \times \sin^2 x$$

Q.8

V.V.I

$$\log z = -\underset{x \to \pi/2}{Lt} \sin x \cos x$$

$$\log z = -(\sin \pi/2 \cos \pi/2)$$

$$\log z = 0$$

$$e^{\log z} = e^0 \qquad [e^{\log z} = z \ \& \ e^0 = 1]$$

$$z = 1$$

Hence limit of the following question is 1.

Q.18 Find the value of

V.V.I

$$\underset{x \to \infty}{Lt} \left(\frac{x+4}{x+2}\right)^{x+3}$$

$$\underset{x \to \infty}{Lt} \left(\frac{x+2+2}{x+2}\right)^{x+3}$$

$$\underset{x \to \infty}{Lt} \left(\frac{x+2}{x+2} + \frac{2}{x+2}\right)^{x+3}$$

$$\underset{x \to \infty}{Lt} \left(1 + \frac{2}{x+2}\right)^{x+3} \qquad [1^\infty \text{ form}]$$

let

$$z = \underset{x \to \infty}{Lt} \left(1 + \frac{2}{x+2}\right)^{x+3}$$

$$\log z = \underset{x\to\infty}{Lt} \log\left(1+\frac{2}{x+2}\right)^{x+3}$$

$$\log z = \underset{x\to\infty}{Lt} (x+3)\log\left(1+\frac{2}{x+2}\right)$$

$$\log z = \underset{x\to\infty}{Lt} \frac{\log\left(1+\frac{2}{x+2}\right)}{\frac{1}{x+3}} \quad \left[\frac{0}{0} \text{ form}\right]$$

Applying L'hospital's rule,

$$\log z = \underset{x\to\infty}{Lt} \frac{\frac{1}{1+\frac{2}{x+2}}\cdot\left(\frac{-2}{(x+2)^2}\right)}{-\frac{1}{(x+3)^2}}$$

$$\log z = \underset{x\to\infty}{Lt} \frac{2}{(x+2)^2} \times \frac{1}{1+\frac{2}{x+2}} \times (x+3)^2$$

$$\log z = \underset{x\to\infty}{Lt} \frac{2}{(x+2)^2} \times \frac{1}{\frac{x+2+2}{(x+2)}} \times (x+3)^2$$

$$\log z = \underset{x\to\infty}{Lt} \frac{2(x+3)^2}{(x+2)(x+4)} \quad \left(\frac{\infty}{\infty} \text{ form}\right)$$

applying L'Hospital's rule

$$\frac{6(x+3)^2}{2x+6} \quad \frac{6(x+3)^2}{2(x+3)} \quad 3(x+3)$$

~~applying L'hospital's rule,~~

$$\cancel{\log z = \underset{x\to\infty}{Lt} \frac{6(x+3)^2}{(x+2)+(x+4)}} \quad \cancel{\left[\frac{\infty}{\infty} \text{ form}\right]}$$

~~applying L'hospital rule~~

$$\cancel{\log z = \underset{x\to\infty}{Lt} \frac{12(x+3)}{2}}$$

$$\log z = \underset{x\to\infty}{Lt} \frac{4(x+3)}{(x+2)+(x+4)}$$

$$\log z = \underset{x\to\infty}{Lt} \frac{4(x+3)}{2x+6}$$

$$\log z = \underset{x\to\infty}{Lt} \frac{\cancel{4}\,2(x+3)}{\cancel{2}(x+3)}$$

$$\log z = \underset{x\to\infty}{Lt}\ 2$$

$$\log z = 2$$

$$e^{\log z} = e^2$$

$$z = e^2 \qquad (e^{\log z} = z)$$

Hence limit of the following question is e^2.

Q. 9.) The value of $\underset{x \to 0}{Lt} \sin x \log x$ is

$\Rightarrow$ Given;

$$\underset{x \to 0}{Lt} \sin x \log x \qquad \left[\sin x = \frac{1}{\operatorname{cosec} x}\right]$$

$$= \underset{x \to 0}{Lt} \frac{\log x}{\operatorname{cosec} x} \qquad \left[\frac{\infty}{\infty} \text{ form}\right]$$

applying L'hospital's rule;

$$= \underset{x \to 0}{Lt} \frac{\frac{1}{x}}{-\operatorname{cosec} x \cdot \cot x}$$

$$= \underset{x \to 0}{Lt} \frac{\frac{1}{x}}{\frac{-\cos x}{\sin^2 x}}$$

$$= -\underset{x \to 0}{Lt} \frac{\sin^2 x}{x \cos x} \qquad \left[\frac{0}{0} \text{ form}\right]$$

applying L'hospital's rule

$$= -\underset{x \to 0}{Lt} \frac{2 \sin x \cos x}{\cos x - x \sin x}$$

$$= 0$$

Hence the limit of this question is 0.

Q.10 $\underset{x \to 0}{Lt} \left[\frac{a^x + b^x + c^x}{3} \right]^{1/x}$

=>

Given;

$$\underset{a \to 0}{Lt} \left[\frac{a^x + b^x + c^x}{3} \right]^{1/x} \qquad [1^\infty \text{ form}]$$

let

$$z = \underset{x \to 0}{Lt} \left[\frac{a^x + b^x + c^x}{3} \right]^{1/x}$$

$$\log z = \underset{x \to 0}{Lt} \log \left[\frac{a^x + b^x + c^x}{3} \right]^{1/x}$$

$$\log z = \underset{x \to 0}{Lt} \frac{1}{x} \log \left[\frac{a^x + b^x + c^x}{3} \right] \qquad \left\{ \log \frac{A}{B} = \log A - \log B \right.$$

$$\log z = \underset{x \to 0}{Lt} \frac{1}{x} \left[\log (a^x + b^x + c^x) - \log 3 \right]$$

$$\left[\frac{0}{0} \text{ form} \right]$$

applying L' hospital's rule

$$\log z = \underset{x \to 0}{Lt} \frac{\left[\frac{1}{a^x + b^x + c^x} \cdot (a^x \log a + b^x \log b + c^x \log c) \right]}{1}$$

$$\log z = \frac{1}{a^0 + b^0 + c^0} \cdot (a^0 \log a + b^0 \log b + c^0 \log c)$$

$$\log z = \frac{1}{1+1+1}(\log a + \log b + \log c)$$

$$\log z = \frac{1}{3}\log(abc)$$

$$\log z = \log(abc)^{1/3}$$

$$e^{\log z} = e^{\log(abc)^{1/3}}$$

$$z = (abc)^{1/3}$$

Hence the limit of the following question is $(abc)^{1/3}$.

Q.11.) Find the maximum value of the function

$$f(x) = \left(\frac{1}{x}\right)^{x}.$$

$\Rightarrow$ Given;

$$f(x) = \left(\frac{1}{x}\right)^{x}$$

let $z = \left(\frac{1}{x}\right)^{x}$

$z = (x^{-1})^{x}$

$z = x^{-x}$

taking log both side

$\log z = \log (x)^{-x}$

$\log z = -x \log(x)$

on differentiating both side

$\frac{1}{z} \cdot \frac{dz}{dx} = -[\log x + 1]$

$\frac{dz}{dx} = -z[1 + \log x]$ $\quad [z = (x)^{-x}]$

$f'(x) = -(x)^{-x}[1 + \log x]$ $\quad \left[\frac{dz}{dx} = f'(x)\right]$

Equating $f'(x) = 0$

$-(x)^{-x}[1 + \log x] = 0$

exponential function can be zero

$\therefore \quad 1 + \log x = 0$

$\log x = -1$

$e^{\log x} = e^{-1}$

$x = e^{-1}$

~~Now, equating $f''(x)$ of~~

~~$f''(x) = f(x)$~~

Now putting the value of $x = e^{-1}$ $f(x)$

$$f(x) = \left(\frac{1}{x}\right)^{x}$$

$$f(e^{-1}) = \left(\frac{1}{e^{-1}}\right)^{e^{-1}}$$

$$f(e^{-1}) = (e)^{1/e}$$

Hence, the maximum value of function is $(e)^{1/e}$.

Q.12. The maximum value of

$f(x) = x^5 - 5x^4 + 5x^3 - 1$ is

$\Rightarrow$ Given;

$$f(x) = x^5 - 5x^4 + 5x^3 - 1$$

equating $f'(x) = 0$

$$f'(x) = 5x^4 - 20x^3 + 15x^2$$

$$5x^4 - 20x^3 + 15x^2 = 0$$

$$5x^2[x^2 - 4x + 3] = 0$$

$$x^2[x^2 - 3x - x + 3] = 0$$

$$x^2[x(x-3) - 1(x-3)] = 0$$

$$x^2[x-1][x-3] = 0$$

$x = 0$ or $x = 1$ or $x = 3$

Now equating $f''(x)$

$$f''(x) = 20x^3 - 60x^2 + 30x$$

at $x = 0$

$$f''(0) = 0$$

at $x=1$

$f''(1) = -10$

at $x=3$

$f''(3) = 540 - 540 + 90$

$f''(3) = 90$

Here $f''(3) = 90 \ (>0)$

$\therefore$ This is minimum value at $x=3$

at $x=1$

$f''(1) = -10 \ (<0)$

$\therefore$ This is maximum value at $x=1$

$f(1) = 1 - 5 + 5 - 1$
$= 0$

The maximum value of this function is 0.

Q.12. Find the maximum and minimum value of

V.V.I. $3x^4 - 2x^3 - 6x^2 + 6x + 1$ in the interval $(0, 2)$.

$\Rightarrow$ Given;

let

$$f(x) = 3x^4 - 2x^3 - 6x^2 + 6x + 1$$

Equating $f'(x) = 0$

$$f'(x) = 12x^3 - 6x^2 - 12x + 6$$

$$f'(x) = 0$$

$$12x^3 - 6x^2 - 12x + 6 = 0$$

$$x = \pm 1, \ 1/2$$

$$x = 1, \ -1, \ +1/2$$

$$x = -1 \notin (0, 2)$$

$$\therefore \ x = 1 \ \& \ 1/2 \in (0, 2)$$

$$f''(x) = 36x^2 - 12x - 12$$

at $x = 1$

$$f''(x) = 36$$

at $x = 1/2$

$f''(1/2) = 9 - 6 - 12$

$= -9$

at $x = 1/2$

$f''(1/2) = -9 \ (< 0)$

$\therefore$ at $x = 1/2$ the function has maximum value

$f''(1) = 36 \ (> 0)$

$\therefore$ at $x = 1$ the function has minimum value

$f(1) = 3 - 2 - 6 + 6 + 1$

$= 2$

$f(1) = 2$

the ~~maximum~~ minimum value is 2

$f(1/2) = \frac{3}{16} - \frac{2}{8} - \frac{6}{4} + \frac{6}{2} + 1 = \frac{39}{16}$

the maximum value is $\frac{39}{16}$.

Q.14. If $f(x)$ and $g(x)$ are respectively e^x and e^{-x} prove that 'c' of cauchy's mean value theorem is the arithmetic mean between a and b.

⇒ Given;

$$f(x) = e^x$$
$$g(x) = e^{-x}$$

interval (a, b)

apply cauchy mean value theorem

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$$\frac{e^{c}}{-e^{-c}} = \frac{e^{b} - e^{a}}{e^{-b} - e^{-a}}$$

$$\frac{e^{c}}{-e^{-c}} = \frac{e^{b} - e^{a}}{e^{-b} - e^{-a}}$$

$$\frac{e^{c}}{-\frac{1}{e^{c}}} = \frac{e^{b} - e^{a}}{\frac{1}{e^{b}} - \frac{1}{e^{a}}}$$

$$-e^{2c} = \frac{e^{b} - e^{a}}{\frac{e^{a} - e^{b}}{e^{a}e^{b}}}$$

$$+e^{2c} = \frac{e^{a}e^{b}(e^{b}-e^{a})}{+(e^{b}-e^{a})}$$

$$e^{2c} = e^{a+b}$$

$$\log e^{2c} = \log e^{a+b}$$

$$2c = a+b$$

$$c = \frac{a+b}{2}$$

Hence which is arithmetic mean.

Q.12. Verify lagrange's mean value theorem for $f(x) = \sin x$ in $[0, \pi]$.

$\Rightarrow$ Given;

$$f(x) = \sin x$$

applying lagrange mean value

$$f'(c) = \frac{f(b) - f(a)}{b-a}$$

$$\cos c = \frac{\sin\pi - \sin 0}{\pi - 0}$$

$$\cos c = 0$$

$c = \pi/2$

$c = \pi/2 \in [0, \pi]$

Hence verified.

Ok